Fiscal Year:	FY 2017	**Task Last Updated:**	FY 10/17/2016
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of II-VI Compound Semiconductors		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:	Feb. 2013 report: Co-Investigator: Dr. Narayanan Ramachandran, Jacobs ESSSA Group, ER42, NASA Marshall Space Flight Center.		
COI Name (Institution):	Ramachandran, Narayanan (Jacobs ESSSA Group)		
Grant/Contract No.:	Internal Project		
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Performance Goal Text:			

Task Description:

Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below.
1a) Vapor Growth

The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source material, transport of vapor species, and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics.

The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS, and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are:

• to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution.

• to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

1b) Melt Growth

The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are:

• to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals.

• to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot.

• to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

• In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities, and defects in the resulting crystal
• Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth

• Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors

• Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments

During this period, we have the following progresses on the categories of melt growth and the vapor growth of ternary II-VI compound semiconductors:
PUBLICATIONS AND PRESENTATIONS:

• An invited presentation, “Effect of impurities on the morphology on electrical properties of multifunctional TAS Crystals: Tl3AsSe3,” was given by Teja Nagaradona, Ching-Hua Su, Michael DeVilbiss, Brad Arnold, Fow-Sen Choa, Lisa Kelly, and N. B. Singh, as an undergraduate research project by T. Nagaradona in Materials Science & Technology 2015 (MS&T15), Columbus, OH, October 4-8, 2015. The ternary TAS crystal has the following potential applications: NLO (frequency conversion) harmonic generator, Acousto-Optic Tunable Filter (AOTF) for MWIR and LWIR hyperspectral imaging, radiation sensor: gamma-ray sensors, and Etalon and prism based MWIR and LWIR imagers. As a recently awarded supporting contract, the presentation reported the accomplishments so far: (1) Single crystals of thallium arsenic selenide (Tl3AsSe3) were successfully grown by Bridgman technique; (2) Arrays of nanocubes were

Task Progress:

observed on the surface during purification; (3) Dielectric constant and resistivity were measured by fabricating small slabs and Tl3AsSe3 has shown clearly higher resistivity in the high purity single crystals.

• A presentation, “Effect of impurities on the morphology on electrical properties of multifunctional TAS Crystals: Tl3AsSe3,” was given by Teja Nagaradona, Ching-Hua Su, Michael DeVilbiss, Brad Arnold, Fow-Sen Choa, Lisa Kelly, and N. B. Singh, as an undergraduate research project by T. Nagaradona in Materials Science & Technology 2015 (MS&T15), Columbus, OH, October 4-8, 2015.

• A presentation, “Double-diffusive convection during growth of halides and selenides,” by N. B. Singh, S.R. Coriell, Ching Hua Su, and W. M. B. Duval, was presented on the International Conference on Multifunctional Materials for Future Applications (ICMFA 2015), Varanasi, India, October 27-29, 2015.

• A paper, “Thermoelectric properties of Tl-doped PbTeSe crystals grown by directional solidification,” by Ching-Hua Su, was published on Journal of Crystal growth. Thermoelectric devices convert heat into electric energy and have been employed in many applications, such as long-duration space exploration missions, renewable energy in automobile, and aerospace industry. With no moving parts, thermoelectric generators have demonstrated the huge advantage of long-duration operational reliability. The conversion efficiency of a thermoelectric material is usually rated by the Figure of Merit, zT = a2sigmaT/k, which is determined by three material properties, i.e., Seebeck coefficient a, electrical conductivity sigma, and thermal conductivity k, and temperature T. The ultimate goal is to improve the overall converting efficiency for the thermoelectric devices to the 15 to 30% range, or zT above 2.0 (depending on operating environment), to be comparable with dynamic power conversion such as Sterling/Brayton cycle. Three p-type Tl-doped PbTe and two Tl-doped PbTeSe crystals were grown by vertical un-seeded directional solidification method. The Tl-doped PbTeSe samples have the highest zT value of 1.63 at 425 to 475°C comparing to the zT value of 1.13 at 410°C for Tl-doped PbTe samples. By substituting 15% of the Te lattice in the Tl-doped PbTe by Se, the Figure of Merit at 450°C range was enhanced by 47% because of the reduction of thermal conductivity by 26% through the mismatched crystalline lattice and, at the same time, an increase of electrical conductivity by about 43% due to the cross-over of light and heavy valence bands at the composition of PbTe0.85Se0.15.

• Two oral presentations were given at SPIE (Defense + Commercial Sensing) conference, Baltimore, MD, April 17-21, 2016. In one presentation “Growth mechanism of nanowires: Binary and ternary chalcogenides,” by N.B. Singh, S.R. Coriell, Ching-Hua Su, R.H. Hopkins, B. Arnold, Fow-Sen Choa, and Brian Cullum, growth of lead selenide nanowires was performed by physical vapor transport method and growth of thallium arsenic selenide nanowire by vapor-liquid-solid (VLS) method. In some cases, very long wires (>1 mm) were formed. The mechanism of nanowire growth from the melt-liquid-vapor medium was presented. In another presentation “Design and growth of novel compounds for radiation sensors: Multinary chalcogenides,” by N.B. Singh, Ching-Hua Su, Teja Nagaradona, B. Arnold, and Fow-Sen Choa, the growth of ternary and quaternary lead and thallium chalcogenides and lead selenoiodides were performed to meet the challenges and demands of low cost and large volume gamma-ray and neutron ray sensors.

• An article, “Method to Improve the Synthesis Process of High-Purity Bulk Multi-Element Compounds,” was published on NASA Tech Briefs, p. 44, Vol.40, No.4, April 2016. The article presents a section of the New Technology Report (NTR1438010053), “Method to Improve Synthesis Process of High-Purity Bulk Multi-Element Compounds,” submitted by Ching-Hua Su. It is claimed, in this invention, that by specifically preparing the fused quartz/silica ampoule and carefully processing the heating procedure the chemical reaction between the sample and the ampoule can be minimized and the crystalline quality of the resultant materials can be vastly improved.

• An article, “Design and growth of novel compounds for radiation sensors: multinary chalcogenides,” by N.B. Singh, Ching-Hua Su, Teja Nagaradona, Brad Arnold, and Fow-Sen Choa, has been published in Proc. SPIE 9824, Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE) Sensing XVII, 982411 (May 12, 2016); doi:10.1117/12.2220405. Increasing threats of radiological weapons have revitalized the researches for low cost large volume gamma-ray and neutron ray sensors. Design and growth of ternary and quaternary lead and thallium chalcogenides and lead selenoiodides for detectors were performed to meet these challenges. Single crystals of several compounds were grown from the melt including Tl3AsSe3, Tl3AsSe3-xSx, TlGaSe2, AgGaGe3Se8, AgxLi1-xAgGaGe3Se8, and PbTlI5-xSex.

• An article, “Growth mechanism of nanowires: Ternary chalcogenides,” by N.B. Singh, S.R. Coriell, R. H. Hopkins, Ching-Hua Su, B. Arnold, Fow-Sen Choa, and Brian Cullum, has been published in Proc. SPIE 9863, Smart Biomedical and Physiological Sensor Technology XIII 986304 (2016), doi:10.1117/12.2220154. Semiconductor nanowires exhibit very exciting optical and electrical properties including high transparency and a several order of magnitude better photocurrent than thin film and bulk materials. Without a catalyst, the mechanism of nanowire growth from the melt-liquid-vapor medium with self-nucleation and uncontrolled growth from different places was observed in the lead selenide (PbSe) and thallium arsenic selenide (Tl3AsSe3) material systems.

• An article, “Effect of pH on the morphology of kidney stones,” Neelesh Agarwal, Stacey Soya, N.B. Singh, Brad Arnold, Fow-Sen Choa, Brian Cullum, and Ching-Hua Su, Proc. SPIE 9863, Smart Biomedical and Physiological Sensor Technology XIII 986303 (2016), doi:10.1117/12.2219939. The process for the formation of kidney stone is very complex phenomena and has some similarity to the crystal growth from a solution. In this study, as a student research project, we have performed experiments using carbonate, oxides and urea to simulate and understand the morphologies of the residue filtered and coarsened in different conditions. We observed that different morphologies of kidney stones can be explained on the basis of acidity and hydration conditions.

• A paper, “Thermal stability of melt grown Tl-doped PbTeSe material for thermoelectric applications,” by Ching-Hua Su, was published in Materials Science in Semiconductor Processing, 56 94-99 (2016). There have been major breakthroughs during the past decade in the enhancement of Figure of Merit for PbTe-based thermoelectric materials. In these reports, the samples were processed by hot pressing, with its different variations, or quench and annealing method. The enhancement has been mainly attributed to the reduction of thermal conductivity through the introduction of different types of nano- or micro-structural inhomogeneities and defects. However, during the practical applications, the thermoelectric devices need to continually operate at elevated temperature with high efficiency for a long period of time. The long-term operation/annealing of these materials will change and modify their microstructure and, consequently, affect their thermoelctric properties. In this report, Tl-doped PbTeSe thermoelectric material has been processed by melt growth of directional solidification, which results in a homogeneous structure of a lower-energy state and, consequently, more thermally stable and mechanically robust. To test its thermal stability, three as-grown and one annealed sample have been characterized. The characterization results show that the annealing at 455°C for 570 h has

essentially no effects on the thermoelectric properties, metallurgical microstructure as well as mechanical hardness of the sample.

• A paper, “Effect of impurities and growth parameters on the quality of Te3AsSe3 optical crystal,” by N. B. Singh, Ching-Hua Su, Teja Nagaradona, Brad Arnold, and Fow-Sen Choa was published on Optical Materials 60, 81-85 (2016). Thallium arsenic selenide (Tl3AsSe3) stoichiometric source materials suitable for crystal growth was synthesized in a horizontal furnace using purified parent components Tl, As, and Se. Single crystal was grown by a two zone vertical Bridgman furnace using capillary for nucleation. A thermal gradient 18–20 K/cm and growth speed of 1 cm/day was used in vertical Bridgman furnace during crystal growth. Crystals were free from micro cracks and precipitates and demonstrated good fabricability. Results indicated that lower gradient and low growth rate are key to reduce thermal stresses in this ternary Tl-As-Se material system. Measured dielectric values in the range of 10 Hz to 100 KHz were almost constant indicating its suitability for applications in a range of frequency where constant dielectric is required.

• A paper, “Optical and morphological characteristics of zinc selenide-zinc sulfide solid solution crystals,” by N. B. Singh, Ching-Hua Su, Brad Arnold, and Fow-Sen Choa was published on Optical Materials 60, 474-480 (2016). The crystal of ZnSe-ZnS solid solution, or ZnSexS(1-x), is one of the selected materials to be grown in the upcoming microgravity flight experiment on ISS, “Crystal Growth of Ternary Compound Semiconductors,” P.I.: Ching-Hua Su. The solid solution crystals have been grown under terrestrial (1-g) condition at MSFC. A detailed optical and morphological characterizations have been performed on the compositions of ZnSe0.91S0.09 and ZnSe0.73S0.27. The purposes of the studies are: (1) to establish the ground-based data for the comparison of the future space-grown crystals and (2) to provide the optical properties of ZnSexS(1-x) crystals for the first time as the bulk crystal growth of this solid solutions is difficult to perform.

SCIENTIFIC COMMUNITY SERVICES:

• Ching-Hua Su has been invited as scientific member of the Editorial Board for the publishing house, Trans Tech Publication (TTP) for the periodicals: Advanced Materials Research (ISSN: 1662-8985) and Materials Science Forum (ISSN: 1662-9752).

• Ching-Hua Su has been invited as a member of Editorial Advisory Board for the journal “Current Smart Materials” of Bentham Science publisher. Current Smart Materials, a peer reviewed journal, publishes reviews, original research articles and thematic issues on experimental and applied research on smart materials. The scope covers contemporary and future smart materials e.g. alloys, polymers, self-healing, shape memory alloys, dielectric elastomers, thermoelectric, magnetocaloric and magnetostrictive, photomechanical, halochromic, chromogenic, piezoelectric, and ferroelectric materials. Coverage also includes applications to sensors and actuators and other smart material systems applied to a range of disciplines.

• Ching-Hua Su has been invited as a member of Editorial Advisory Board for the journal “Recent Patents on Materials Science” of Bentham Science publisher. Recent Patents on Materials Science publishes review and research articles, and guest edited thematic issues on recent patents in all areas of materials science. A selection of important and recent patents on materials science is also included in the journal. The journal also covers recent research (where patents have been registered) in fast emerging areas of materials science, biomaterials, biological materials, magnetic materials, medical implant materials, nanomaterials, ceramics, plastics, polymers, semi-conductors, coatings, composites, paper, wood, textile, methods, design, and techniques.

• Ching-Hua Su has served as a member of the International Advisory Board for the conference, “INTERNATIONAL CONFERENCE ON MULTIFUNCTIONAL MATERIALS FOR DEVICE APPLICATION (ICMDA-2016),” to be held in Patna, India from October 26 to 28, 2016. The objective of the conference is to discuss and review the developments in synthesis, characterization and application of bulk, thin film, quasi phase materials, nanocrystals and nanowires, organic and inorganic materials for commercial, health, energy, nuclear, and military systems as well as its applications in industry academia and various devices.

• Ching-Hua Su has completed the technical reviewing and panel discussions (through teleconference) of two proposals submitted in response to the NASA Headquarters (HQ) solicitation of Flight Opportunities SpaceTech–REDDI–2016. He served as primary reviewer on one proposal, i.e., collecting and compiling comments from five reviewers, leading the discussions, and writing the final summary as well as served as secondary reviewer on the other proposal by providing the evaluation comments and participating in discussions.

• By the request of the editors, Ching-Hua Su has performed technical reviews on manuscripts submitted to “Journal of Phase Equilibria and Diffusion” and “Journal of Physics and Chemistry of Solids.”

STUDENTS’ SUPPORTS AND TRAINING:

To support NASA Education and Public Outreach Program, the flight experiment’s contracts with Stony Brook University and University of Maryland, Baltimore County not only has performed defects characterization on the grown samples but also has been training students for the future workforce especially for NASA.

• A graduate student of Stony Brook University, Ouloide Yannick Goue, has been awarded a doctor of philosophy degree in materials science and engineering with a Ph.D. dissertation title of “Characterization of Structural Defects in Wide Band-Gap Compound Materials for Semiconductor and Opto-Electronic Applications.” Single crystals of CdZnTe have been grown in Marshall Space Flight Center (MSFC) under controlled Cd-overpressure to reduce the Te inclusions so as to improve the detection capability for the applications as advanced room-temperature X-ray, Gamma-ray detectors. The grown crystals characterized in Stony Brook University by SXRT (Synchrotron X-Ray Topography) revealed that under certain Cd overpressures and growth conditions, the quality of the grown boule improved. Similarly, transmission infrared (IR) microscopy showed a correlation between the size/density and distribution of Te inclusions/precipitates and Cd overpressure.

• The following students at University of Maryland, Baltimore County, have been involved and learning different skills in this program:

Stacey Sova (Graduate student registered for Ph.D. degree); Working on spectral characteristics of the sample (Near IR and LWIR characteristics) ; Eric Languirand (Graduate student, registered for Ph.D. degree): Working on defect characterization by Raman spectra ; Sara Bohorfoush (Senior, Registered for BS degree): Working on photoluminescence spectral characteristics of the material ; Sonali Saraf (Junior, Registered for BS, degree Double

	major); Working on X-ray and morphological characterization by Scanning electron microscope and surface modification ; Jayati Bhavsar (Sophomore, Registered for BS): Working on developing surface fabrication and etchants to improve the surface of the material.
Bibliography Type:	Description: (Last Updated: 02/18/2025)
Articles in Other Journals or Periodicals	Marshall Space Flight Center. "Method to Improve the Synthesis Process of High-Purity Bulk Multi-Element Compounds. The article presents a section of the New Technology Report (NTR1438010053), 'Method to Improve Synthesis Process of High-Purity Bulk Multi-Element Compounds,' submitted by Ching-Hua Su." NASA Tech Briefs. 2016 Apr 1. http://www.techbriefs.com/component/content/article/ntb/tech-briefs/manufacturing-and-prototyping/24253 ; accessed 10/17/16. , Apr-2016
Articles in Peer-reviewed Journals	Su C-H. "Thermal stability of melt grown Tl-doped PbTeSe material for thermoelectric applications." Materials Science in Semiconductor Processing. 2016 Dec;56:94-9. http://dx.doi.org/10.1016/j.mssp.2016.07.024 , Dec-2016
Articles in Peer-reviewed Journals	Singh NB, Su C-H, Arnold B, Choa F-S, Nagaradonaa T. "Effect of impurities and growth parameters on the quality of Tl3AsSe3 optical crystal." Optical Materials. 2016 Oct;60:81-5. http://dx.doi.org/10.1016/j.optmat.2016.07.009 , Oct-2016
Articles in Peer-reviewed Journals	Singh NB, Su C-H, Arnold B, Choa F-S. "Optical and morphological characteristics of zinc selenide-zinc sulfide solid solution crystals." Optical Materials. 2016 Oct;60:474-80. http://dx.doi.org/10.1016/j.optmat.2016.08.031 , Oct-2016
Articles in Peer-reviewed Journals	Su C-H. "Thermoelectric properties of Tl-doped PbTeSe crystals grown by directional solidification." Journal of Crystal Growth. 2016 Apr;439(1):80-6. http://dx.doi.org/10.1016/j.jcrysgro.2016.01.012 , Apr-2016
Papers from Meeting Proceedings	Singh NB, Su C-H, Nagaradona T, Arnold B, Choa F-S. "Design and growth of novel compounds for radiation sensors: multinary chalcogenides." Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE) Sensing XVII, Baltimore, Maryland, April 17, 2016. Proceedings SPIE 9824, Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE) Sensing XVII. Volume 982411 (May 12, 2016); http://dx.doi.org/10.1117/12.2220405 , May-2016
Papers from Meeting Proceedings	Singh NB, Coriell SR, Su C-H, Hopkins RH, Arnold B, Choa F-S, Cullum B. "Growth mechanism of nanowires: binary and ternary chalcogenides." Smart Biomedical and Physiological Sensor Technology XIII, Baltimore, MD, April 17, 2016. Proceedings of SPIE 9863, Smart Biomedical and Physiological Sensor Technology XIII. Volume 986304 (May 13, 2016). http://dx.doi.org/10.1117/12.2220154 , May-2016
Papers from Meeting Proceedings	Agarwal N, Soya S, Singh NB, Arnold B, Choa F-S, Cullum B, Su C-H. "Effect of pH on the morphology of kidney stones." Smart Biomedical and Physiological Sensor Technology XIII, Baltimore, MD, April 17,2016. Proceedings of SPIE 9863, Smart Biomedical and Physiological Sensor Technology XIII. Volume 986303 (May 13, 2016). http://dx.doi.org/10.1117/12.2219939 , May-2016